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A multi-expert deep learning framework with LLM-guided arbitration for multimodal histopathology prediction☆

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ABSTRACT

Recent advances in deep learning have significantly improved the accuracy of computational pathology; however conventional model ensembling strategies often lack adaptability and interpretability hindering the clinical adaptability. While multiple artificial intelligence (AI) expert models can provide complementary perspectives, simply aggregating their outputs is often insufficient for handling inter-model disagreement and delivering interpretable decisions. To address these challenges, we propose a novel multi-expert framework that integrates diverse vision-based predictors and a clinical feature-based model, with a large language model (LLM) acting as an intelligent arbitrator. By leveraging the contextual reasoning and explanation capabilities of LLMs, our architecture dynamically synthesizes insights from both imaging and clinical data, resolving model conflicts, and providing transparent, rational decisions. We validate our approach on two cancer histopathology datasets, namely the HMU-GC-HE-30K which is a gastric cancer dataset containing pathology images only, and the BCNB which is a breast cancer biopsy dataset that is multimodal — contains pathology imaging and clinical information. Our proposed multi-expert, LLM arbitrated framework (MELLMA) outperforms convolutional neural networks (CNNs), and transformers, which are currently the de facto and state-of-the-art classification ensemble models, with better overall results. We test different LLMs as arbitrators, namely LLaMA, GPT variants, and Mistral. Further, our proposed framework outperforms strong single-agent CNN/ViT baselines on the datasets, and ablations show that learned per-agent trust materially improves the arbitrator's decisions without altering prompts or data. These experimental results demonstrate that LLM-guided arbitration consistently provides more robust and explainable performance than individual models, conventional ensembling with majority vote, uniform average, and meta-learners. The results obtained highlight the promise of LLM-driven arbitration for building transparent and extensible AI systems in digital pathology.

1. Introduction

Advancements in digital pathology have been revolutionized by artificial intelligence (AI) techniques, with the inclusion of automated tools for disease diagnosis, subtyping, and outcome prediction. During the past decade, deep learning models, in particular convolutional neural networks (CNNs) (LeCun et al., 2015) and vision transformer (ViT) (Dosovitskiy et al., 2020) models, have significantly advanced accuracy and efficiency in computational pathology. Despite this success, designing a single all-powerful model which can handle the diversity of patterns in histopathology images remains challenging due to the complexity and heterogeneity of tissue morphology. For example, hematoxylin and eosin (H&E) stained images contain many different types of tissues and staining variations — no single model may excel uniformly in all such contexts. This motivates us to explore multiple specialized models jointly so that each can contribute where it performs the best. Recent research suggests that the use of "experts" trained in different tissue characteristics can improve performance by focusing on specific intratumoral tissue subtypes (Wu et al., 2025).

Ensembling in deep learning is a popular approach that can leverage the strengths of multiple models. By combining predictions from an array of classifiers, ensembles often show more robust and accurate results than any single model in medical imaging (Supriyadi et al., 2025). However, naive ensemble strategies, such as the majority vote or average of outputs have limitations. Traditional static ensembles treat

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each component model equally and lack awareness of when a particular model is likely to be correct or not. Consequently, ensembles can be underconfident or overreliant on poorly performing models, especially if some models consistently misclassify certain tissue types (Müller et al., 2022). Recent work has begun to address this issue by dynamically weighting or selecting models per sample (Vignesh et al., 2024). Salvi et al. (2024) propose an uncertainty-guided ensemble that selects the two most reliable models for each image based on Monte Carlo dropout uncertainty, outperforming fixed ensembles. Techniques such as mixture-of-experts (MoE) with trainable gating networks have also been applied in pathology, allowing specialized models to handle heterogeneous tissue patterns in a slide (Wu et al., 2025). These approaches underscore the value of an intelligent arbitration mechanism within an ensemble, rather than a one-size-fits-all fusion.

Another critical consideration in medical AI is interpretability and trust from clinicians. Typical deep learning ensembles operate as black boxes offering predictions without explanations, which is often insufficient in high-stakes domains like cancer diagnosis. Clinicians are more likely to adopt AI assistance if the system can justify its decision in human-understandable terms (Huang et al., 2025a). Recent advances in agentic AI (Zou and Topol, 2025; Gorenshtein et al., 2025) and large language models (LLMs) (Yu et al., 2025) suggest a pathway toward more explainable and flexible AI systems in healthcare. LLMs can integrate diverse inputs and produce fluent textual explanations, effectively serving as a "brain" that reasons over the outputs of other tools or models. Pioneering frameworks like AURA (Fathi et al., 2025) and MMedAgent (Li et al., 2024) have demonstrated the promise of combining multiple specialized models with an LLM-based coordinator, enabling dynamic decision-making and explanations in medical image analysis.

In this study, we propose a multi-agent decision framework in which multiple specialized models ("agents") independently analyze the same histopathological input from different representational perspectives. Their outputs are then adjudicated by an LLM acting as an intelligent arbitrator, capable of dynamically weighting agent opinions based on case-specific trust metrics. This arbitration is not limited to image-derived predictions; the framework is extensible to include agents trained on non-image clinical data, thus more closely emulating a pathologist's multimodal reasoning process. We validate our framework in two complementary datasets - (1) HMU-GC-HE-30K – a large scale hematoxylin and eosin (H&E) stained histopathology dataset containing image data (Lou et al., 2025). This serves as a baseline for evaluating our proposed framework in a purely imaging-specific context, where multiple CNN and transformer agents contribute to the decision process. (2) BCNB – a breast cancer histopathology dataset that includes both H&E image patches and detailed patient-level clinical metadata (Xu et al., 2021). Unlike prior learned ensembles or MoE gates (Supriyadi et al., 2025), we introduce a new learned per-expert reliability (trust) estimator whose output is explicitly consumed by the LLM via a structured prompt/rubric, allowing trust-conditioned arbitration and succinct, auditable justifications. In what follows we use the term *expert* to refer to a trained vision model (capability view), and the term *agent* when the same model is surfaced to the arbitrator through a lightweight interface (coordination view) that standardizes its predicted class, calibrated confidence, and (where applicable) clinical cues. Thus, both terms denote the same module from different perspectives.

2. Related works

Ensembling multiple models is a well-established technique in medical imaging to increase robustness and accuracy beyond what a single classifier can achieve. The inherent heterogeneous features of histological tissue types have motivated the use of specialized expert models with each proficient in certain tissue subtypes, combined into an ensemble of models. For example, Wu et al. (2025) proposed a mixture-of-experts framework in which different neural networks focus on distinct intratumoral morphologies; a gating network then selects or weighs these "experts" for each image, which improved performance on gigapixel whole-slide images. Similarly, Hashimoto et al. (2024) integrated histology image models with a flow cytometry expert in a gated ensemble for lymphoma classification, and demonstrated that this multimodal Mixture-of-Experts (MoE) outperformed both any single modality model and a naive fusion of modalities. Their gated approach achieved higher accuracy, 72.3% versus the approximately 70% accuracy of the baselines, and provided interpretable visualizations of the regions and cellular features on which each expert relied. These successes illustrate that the intelligent combination of multiple predictors can better handle the complexity of pathology data than one-size-fits-all models.

In digital pathology, early ensemble strategies often used rule-based fusion (e.g., majority voting or averaging of model outputs) (Majumdar et al., 2023; Xiao et al., 2018; Yuan et al., 2025). Such static schemes assume that each model is equally reliable and thus cannot adapt when an individual model underperform. This can lead to suboptimal results; for instance, an ensemble might overly rely on a poor model or dilute the contribution of a highly accurate one. To address these issues, researchers have developed learned ensembling methods that dynamically adjust to each case. Salvi et al. (2024) propose an uncertainty-guided ensemble that, for each input, selects the two most reliable segmentation models based on Monte Carlo dropout confidence, thereby outperforming a fixed average of all models. Similarly, trainable gating networks in mixture-of-experts models allow the ensemble to learn which expert to trust for a given input, rather than treating all experts equally (Wu et al., 2025; Mu and Lin, 2025; Li et al., 2023). Despite their superior performance, these learned or weighted ensembling techniques typically operate as black boxes. They produce a final prediction without explaining the basis for choosing one model over another. In critical domains such as cancer diagnosis, lack of transparency remains a major drawback. Clinicians are more inclined to trust and adopt AI if the system can justify its decision with human-understandable reasoning (Nasarian et al., 2024; Chaddad et al., 2023). Conventional ensembles and meta-learners, however, offer little insight into why a particular model was favored in a given case, undermining user trust.

Recent advances in large language models (LLMs) have opened new avenues to make AI decision making more adaptive and interpretable in the clinical context. LLMs possess powerful reasoning and natural language generation capabilities, enabling them to serve as high-level "brains" on top of traditional models (Cheng, 2024). Several works have begun exploring this paradigm. Fathi et al. (2025) introduced AURA, an agentic framework where a multimodal LLM (Qwen-32B) orchestrates a toolkit of image analysis modules (segmentation, counterfactual generation, etc.) and provides dynamic visual-linguistic explanations for medical images. Another notable example is MMedAgent, proposed by Li et al. (2024). MMedAgent is an LLM-driven multimodal medical agent that can call upon a set of expert models ("tools"), which span imaging, text, and other data to solve a given task. Through instruction tuning, the LLM learns to choose the appropriate specialist models for each query and to integrate their outputs. Crucially, it also demonstrated the flexibility of an LLM-based controller to incorporate new tools and update its decisions, an adaptability hard to match with conventional learned ensembles. Also there is an emergence of LLM-powered diagnostic assistants that operate directly on histology images. Lu et al. (2023) introduced PathChat, a vision language foundational model trained on over 100 million pathology images and extensive image caption pairs. PathChat combines a custom image encoder with a large language model, allowing it to answer open-ended questions about a pathology image achieving about 87% diagnostic accuracy on a diverse set of cases. Similarly, other multimodal LLM systems have been tailored for pathology; for example, a PathAsst model (Sun et al., 2024) that couples a pathology specific

CLIP vision encoder with a Vicuna LLM to generate pathology reports, and the Quilt-LLAVA (Seyfioglu et al., 2024) model that uses over 100k Q&A pairs to enable complex diagnostic reasoning across multiple image patches. In a broader clinical AI context, researchers have even employed multiple LLMs in a consensus building conversation to improve diagnoses. Ke et al. (2024) simulate a team of AI "clinicians" with distinct roles (e.g. a devil's advocate to challenge biases, a facilitator, etc.) all implemented via GPT-4 agents. Through their multi-agent debate, an initially biased diagnosis was often corrected, raising the accuracy of the final consensus diagnosis dramatically from 0% to over 71% in their study.

Despite the advances described above, there remains a clear gap at the intersection of ensemble modeling and LLM-driven reasoning in digital pathology. Traditional ensembling methods, whether rule-based or learned, generally improve accuracy but do not provide transparency or case-specific adaptability in their decision logic. In contrast, recent LLM-centric approaches demonstrate interpretability and flexible integration of multimodal data (Hager et al., 2024). However, so far they have largely been applied either to single model outputs or to broad multi-tool settings, rather than resolving conflicts among multiple experts within the same domain. In other words, to the best of our knowledge, no prior work has fully harnessed reasoning ability of an LLM to arbitrate between competing model predictions in pathology using explicit trust or confidence cues from those models. Our work addresses this unmet need by integrating the performance benefits of a dynamic ensemble with the interpretability of an LLM. We introduce a multi-expert LLM arbitration (MELLMA) framework in which multiple high-performing CNN and transformer based classifiers (the "experts") first independently analyze histology patches, and a dedicated meta-learner estimates each model's reliability (or "trustworthiness") for the given case. An LLM then serves as the arbiter which receives the models' predictions, confidence scores, and trust estimates, and using strong prompt engineering reasons to produce a final diagnosis. The LLM arbitrator can weight the opinions of each models based on its trust and general agreement, mimicking an expert pathologist who listens to colleagues but gives more credence to those with a strong track record on the specific problem at hand. This LLM-driven arbitration not only improves decision accuracy by dynamically selecting the right model for the right situation, but also provides an auditable rationale in natural language for each decision. By combining quantitative trust weighting with qualitative explanation, our approach fills the critical gap between black-box ensembling and explainable AI in digital pathology.

Our framework departs from conventional ensemble designs in three key ways: (1) **Dynamic arbitration** – Rather than averaging predictions, the LLM arbitrator selectively prioritizes agent outputs depending on the level of agreement between models, confidence scores, and learned trust patterns. (2) **Modality integration** – The system accommodates both image-derived and structured clinical features, allowing richer and more clinically grounded decision making. (3) **Interpretability through reasoning traces** – The LLM arbitrator can generate concise natural language logics explaining why the predictions of certain agents were favored, helping transparency and clinical acceptance. Building on these design departures, our key contributions are as follows.

- A multi-agent decision making framework: We utilize MELLMA for histopathology image analysis, integrating both CNN and transformer models as image-based agents and extending to include non-image clinical data.
- Trust aware LLM arbitration: We employ a compact, case conditional reliability estimator (feed forward gating network) to produce per case, and per agent trust scores. We expose that trust vector directly to the LLM via a structured prompt and decision rubric, enabling trust-conditioned arbitration with auditable reasons, an under explored design choice in digital pathology.
- An LLM-based arbitration: Our arbitration mechanism dynamically reconciles multiple agent predictions using confidence and trust measures, producing both final decisions and natural language explanations.
- Evaluation on two complementary datasets: We test our MELLMA approach on two challenging pathology datasets, namely the HMU-GC-HE-30K (imaging only) (Lou et al., 2025) and the BCNB (imaging + clinical) (Xu et al., 2021). Our framework demonstrates adaptability across single and multi modal contexts with evidence of improved robustness over conventional ensemble strategies, particularly in ambiguous or clinically challenging cases.

3. Methods

Our MELLMA framework comprises multiple specialized image analysis models ("experts"), a meta-learner that assigns adaptive trust scores to experts on a per-case basis, and a Large Language Model (LLM) that arbitrates the final decision. Our overall model is depicted in Fig. 1. Given a histopathology input, patch from a whole slide image (WSI), each expert produces a probability distribution over the target classes and an associated confidence. The meta-learner reads per agent summary features (predicted class and confidence) and outputs per agent trust scores for the current case. A structured prompt then composes (i) each expert's predicted class and confidence, (ii) the meta-learner's trust scores, and (iii) any available clinical features. The LLM reasons over this summary and returns the consolidated final class (and a short rationale). This multi-agent, multi-model pipeline leverages complementary visual features from diverse experts and the LLM's reasoning to improve robustness and interpretability.

3.1. Multi-expert image models ("Agents")

We instantiate M image *experts* (e.g. CNN/ViT backbones) and expose each through a thin *agent* wrapper that presents a stable interface to the arbitrator: model ID, top class $\hat{y}_j$, calibrated confidence c_j, and the class probability vector $\mathbf{p}_j$. In this sense, the "expert" captures the learned capability, while the "agent" describes the same module's role in the coordination protocol. Formally, each expert function E_j emits a class-probability vector p_j and a scalar confidence score c_j. Let there be M experts and K classes. For an input patch x, the jth expert E_j produces $p_j(x) = [p_j(y=1), \dots, p_j(y=K)]$ and $\hat{y}_j = \arg\max_k p_j(y=k)$ with corresponding confidence $c_j = \max_k p_j(y=k)$. For an input image I, the jth expert emits a class-probability vector,

$$\mathbf{p}_j \;=\; E_j(I) \in [0,1]^K \quad \text{with} \quad \sum_{k=1}^{K} p_j(y=k \mid I) = 1,$$

and a scalar confidence,

$$c_j \;=\; \max_{1\le k\le K} \; p_j(y=k \mid I).$$

Experts use heterogeneous architectures (e.g., a CNN emphasizing local textures vs. a transformer capturing global context), promoting diversity. Each expert's top class prediction is $\hat{y}_j = \arg\max_k p_j(y = k \mid I)$.

We use two CNNs (ResNet-50, EfficientNet-B3) and two vision Transformers (ViT-B/16, Swin-V2-S) to maximize architectural diversity while keeping model size and compute modest. CNNs provide strong local texture inductive bias, whereas transformers capture longer range contexts. All four use the standard supervised ImageNet-1K pretrained weights released by the original model authors, and are widely adopted in computational pathology, improving reproducibility (Boudissa et al., 2024; Debsarkar et al., 2025). We deliberately avoided much larger backbones (e.g., Swin-L/ConvNeXt-XL) so that the results reflect the arbitration mechanism rather than the raw backbone capacity. Pilot checks with larger variants showed the same relative arbitration gains.

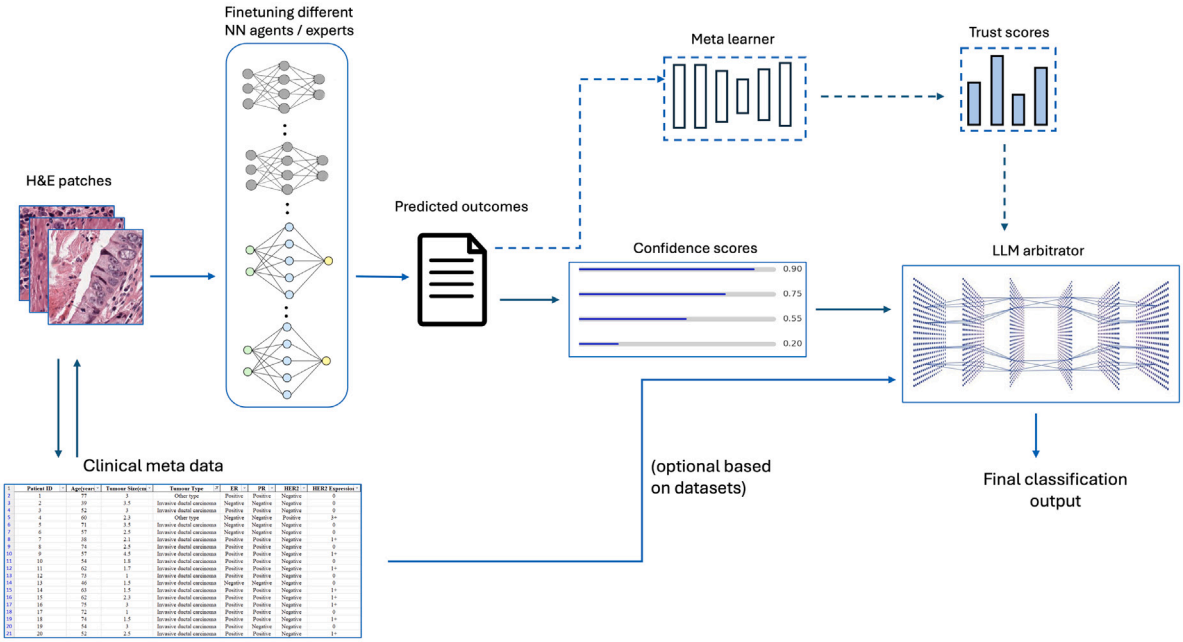


Fig. 1. Overview of the multi-expert, LLM arbitrated (MELLMA) framework. For each patch, M vision agents (e.g., ResNet-50, ViT-B/16, EffNet-B3, SwinV2-S) produce a predicted class $\hat{y}_j$ and confidence c_j. A small meta-learner outputs a trust score τ_j per agent; on BCNB, a compact clinical summary s_{clin} is also available. These components are assembled into a structured prompt consists of an agent table $\{\hat{y}_j, c_j, \tau_j\}_{j=1..M}$, optional s_{clin}, and a fixed decision rubric. These are then sent to the LLM arbitrator (GPT-4/4o, LLaMA-3, Mistral-7B), which returns a JSON decision `{final_class, reason}`. Dashed elements indicate the TRUST OFF ablation, where τ_j and trust referencing rubric text are removed. Patch level outputs can be aggregated to the patient level by mean probability pooling. Throughout, "agent" denotes an expert model presented via the arbitration interface.

We selected four mid-sized, widely used ImageNet-1K pretrained backbones namely ResNet–50, EfficientNet–B3, ViT–B/16, and SwinV2-S to provide architectural diversity while avoiding both overfitting and excessive computational burden. Smaller models such as ResNet-18 generally offer weaker representational capacity and less stable predictions in histopathology tasks, while substantially larger backbones (e.g., ResNet–101, EfficientNet–B4/B5, ViT–Large, Swin-Large) increase memory footprint and training cost without yielding consistent improvements on datasets of this scale. Using more than four experts produced diminishing returns and increased computational overhead, whereas fewer experts reduced the diversity of complementary features available for arbitration. The chosen four-expert configuration therefore provides a balanced and practical trade-off between representational diversity, robustness, and throughput for LLM arbitration.

The chosen backbones strike a balance between architectural diversity and practical feasibility. Preliminary checks with larger variants (e.g., ResNet-101, EfficientNet-B4/B5, ViT-Large, Swin-Large) did not yield measurable gains on either dataset and often showed early signs of overfitting due to the moderate dataset sizes and class imbalance, particularly in BCNB. We therefore adopt mid-sized, widely used backbones with ImageNet-1K initialization, which provide complementary inductive biases (local texture cues, hybrid CNN/attention processing, global self-attention, hierarchical transformer reasoning) without excessive model capacity. This selection provides diverse and stable expert signals for arbitration while keeping computation tractable.

All four expert models were initialized from publicly available ImageNet-1K pretrained weights. Each model receives 224 × 224 H&E patches as input after standard normalization. During training, we applied commonly used augmentations for histopathology, including random horizontal and vertical flips, small rotations, color jitter, and mild stain perturbations. ResNet-50 and EfficientNet-B3 were optimized using AdamW with a cosine learning-rate schedule, while ViT-B/16 and Swin-V2-S used AdamW with layerwise learning-rate scaling following standard practice for transformer backbones. Training was performed for 30–50 epochs with early stopping based on validation loss, and patient-level splits were enforced to avoid data leakage. These models were chosen because they represent complementary architectural families (CNN, CNN-hybrid, ViT, hierarchical transformer), providing diverse feature representations for the LLM arbitration stage.

Recent pathology foundation models can serve as strong visual encoders, but they typically require task-specific fine-tuning and involve training procedures and output formats that vary widely across model families. Since our goal in this study is to evaluate the proposed arbitration mechanism under a controlled and comparable setting, we use four widely adopted ImageNet-1K classifiers with consistent training pipelines. The MELLMA framework is fully flexible, and future versions can incorporate foundation models whenever they are fine-tuned for the target task and provide outputs that can be combined through arbitration.

3.2. Meta-learner for adaptive trust weighting

Rather than naive voting or averaging, we employ a meta-learner, a conventional feed forward gating network, that predicts a per-expert trust distribution for each case (one score per agent that sums to one). The meta-learner ingests a fixed length vector summarizing the experts' outputs (predicted class and confidence for each agent and returns trust scores via a compact MLP: *Linear 64* → *ReLU* → *Linear 64* → *ReLU* → *Linear M* → *Softmax*, where M is the number of agents (see Fig. 2).

We encourage high trust on experts who are correct for a given sample by maximizing the log-probability assigned to the set of correct experts. Concretely, with $\tau^{(i)} = \mathrm{softmax}(g_\theta(z^{(i)}))$ we minimize,

$$\mathcal{L}(\theta) \;=\; -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M}\mathbf{1}\big[\hat{y}_j^{(i)} = y^{(i)}\big]\;\log \tau_j^{(i)}\,, \tag{1}$$

where $\mathbf{z}$ denotes the per-agent summary features and g_θ is the MLP. The loss increases trust on agents that are correct for that case, that is, it maximizes the log trust assigned to agents whose prediction matches the ground truth. Here, N denotes the number of training patches and

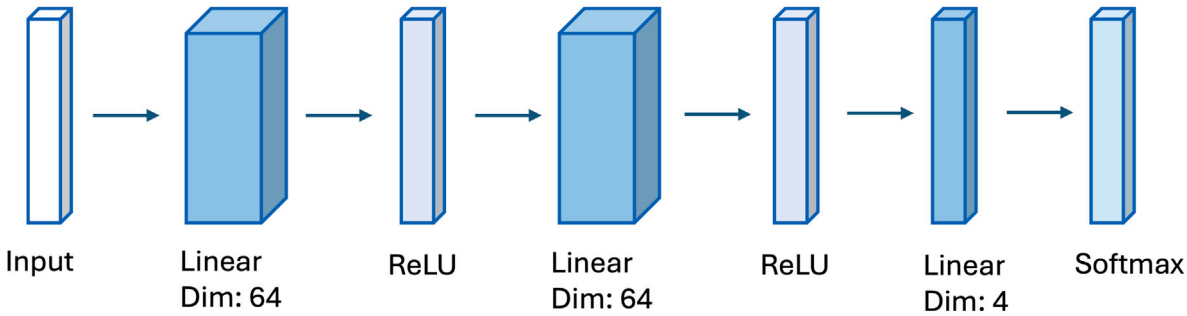


Fig. 2. Meta-learner (trust estimator). Architecture of the meta-learner used for adaptive trust estimation. The model is a multilayer perceptron that ingests agent level features (predicted class and confidence), applies two 64 unit fully connected layers with ReLU activations, and ends with a fully connected layer whose size equals the number of agents. A softmax layer converts these outputs into a normalized trust distribution.

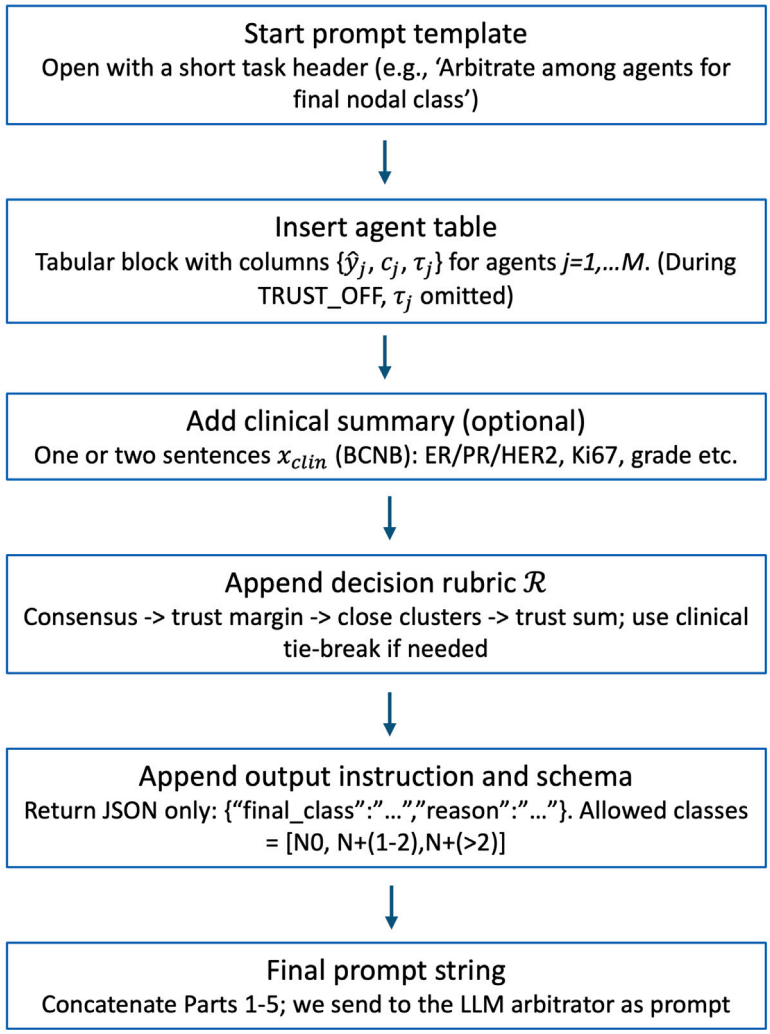


Fig. 3. Prompt schematic for LLM arbitration. The prompt lists each expert's class and confidence, meta-learner trust scores, and for BCNB only a compact clinical summary. A concise rubric specifies how to weigh consensus, dominant trust, near-ties, and trust sum, with clinical evidence as tie-break. The LLM returns a JSON object with the final class and a brief, human-readable reason.

M is the number of experts. Index $i \in \{1, \dots, N\}$ runs over patches, and $j \in \{1, \dots, M\}$ runs over experts. For patch x_i with ground-truth label $y^{(i)}$, expert E_j predicts a top-1 class $\hat{y}_j^{(i)}$, and the meta-learner assigns a trust value $\tau_j^{(i)}$ to the (i, j) pair. The indicator function $\mathbf{1}\left[\hat{y}_j^{(i)} = y^{(i)}\right]$ equals 1 only when expert E_j is correct on patch x_i and 0 otherwise. Thus, this objective increases the log-trust on experts that match the ground-truth label for each sample.

Given trust scores, a calibrated ensemble distribution is given by,

$$P_{\text{ens}}(y = k \mid I) = \sum_{j=1}^{M} \tau_j \, p_j(y = k \mid I), \qquad k = 1, \dots, K. \tag{2}$$

This MoE style weighting allows the pipeline to rely more on the expert(s) best suited for a particular morphology. We consider two utilizations of trust: meta-pick, which selects the single agent with the

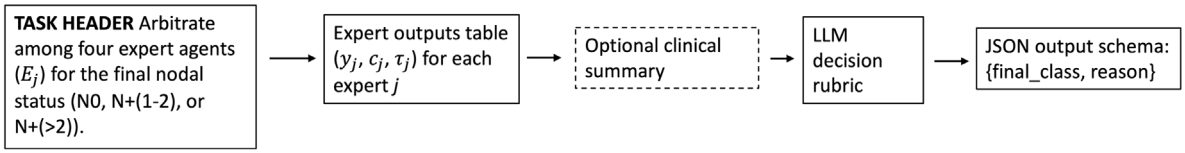


Fig. 4. Simplified overview of the structured prompt. The LLM receives (i) a table of expert predictions (class, confidence, trust), (ii) an optional clinical summary (dashed block), and (iii) a decision rubric with a JSON schema. Here, y_j denotes the predicted class from expert E_j, c_j denotes the corresponding confidence score, and τ_j is the trust value assigned to expert E_j by the meta-learner.

highest trust for the final class (used as a baseline in experiments), and meta-ens, which uses the trust-weighted mixture P_{ens}. To isolate the value of arbitration, we consider these two non-LLM fusions: (i) *meta-pick* - predict the class from the single expert with the highest trust score; and (ii) *meta-ensemble* - form the trust-weighted mixture $P_{\text{ens}}(y) = \sum_j \tau_j \, p_j(y)$ and predict $\arg\max_y P_{\text{ens}}(y)$.

The meta-learner receives a low-dimensional input composed of each expert's predicted class, confidence value, and optional clinical features. We therefore adopt a lightweight MLP with two hidden layers of width 64, which we found to be sufficient for modeling the reliability patterns across experts. Preliminary experiments with wider or deeper architectures (e.g., 128–128–M and 256–256–M) did not improve validation accuracy and sometimes led to overfitting due to the limited input dimensionality. The 64–64–M configuration provided stable convergence and the best trade-off between expressiveness and generalization. We also experimented with classical machine-learning models (logistic regression, random forests, gradient-boosted trees, and support vector machines) using the same expert-level feature inputs. These alternatives consistently underperformed and exhibited less stable validation behavior, reinforcing the use of the lightweight 64–64–M MLP as the trust estimator.

3.3. Clinical data integration

For the BCNB dataset, per-case clinical variables are available (e.g., age, tumor size/grade, ER/PR/HER2, etc.). We integrate them in two places.

(1) Trust estimation. We extend the meta-learner input to include clinical features $\mathbf{x}_{\text{clin}}$, resulting in

$$\tau = M([\mathbf{z}, x_{\text{clin}}]),$$

where $\mathbf{z}$ denotes the per-agent summary features derived from the experts' outputs (predicted class and confidence).

(2) LLM context. We also pass a compact clinical summary into the LLM prompt. This mirrors clinical reasoning, where human experts weigh imaging with patient context, while requiring no additional training of the LLM. The clinical block is optional and does not affect the arbitration mechanism. When clinical metadata are sparse or weakly predictive, the LLM arbiter naturally relies more heavily on expert image-based predictions and the trust vector. Because HMU–GC–HE–30K does not contain paired clinical metadata, multimodal prompting can be evaluated only on BCNB. Accordingly, HMU is used to benchmark the image-only configuration of MELLMA, while BCNB demonstrates how the same arbitration mechanism incorporates clinical cues when they are available. Since only BCNB provides paired clinical metadata, multimodal prompting can be evaluated exclusively in this dataset. The HMU–GC–HE–30K dataset naturally benchmarks the image-only configuration of the MELLMA framework, while BCNB demonstrates how the same arbitration mechanism incorporates clinical cues when such information is available.

3.4. LLM arbitration with structured prompting

Instead of fixing the final class via a rigid rule, we delegate the final decision to an LLM acting as an arbitrator. We construct a structured prompt that includes: (i) a table of experts with predicted class and confidence c_j; (ii) the meta-learner's trust scores τ_j; (iii) a brief clinical summary when available; and (iv) a precise instruction to return only a JSON object with the final class and a short reason. The LLM's reply is parsed to extract the class and rationale.

To guide consistent behavior, the prompt encodes a *decision rubric* that prioritizes: (i) consensus among experts when all are both confident and trusted; (ii) an expert with a dominant trust margin; (iii) agreement clusters with tight trust spread; and (iv) overall trust sum across classes. Clinical cues (BCNB only) serve as tie-break evidence when model signals are ambiguous, e.g., higher tumor burden indicators (larger size, LVI+, higher grade, ER/PR-, HER2+, high Ki67, younger age) bias toward heavier nodal involvement, whereas the opposite leans conservative.

In Fig. 3, we show a detailed workflow for our general prompt development. We construct the prompt in three main parts. (i) *Agent table:* for each vision agent j, we list its predicted class $\hat{y}_j$, confidence c_j, and trust score τ_j (note: τ_j is omitted in TRUST OFF). (ii) *Optional clinical summary:* a 1–2 sentence s_{clin} (BCNB only). (iii) *Rubric + output schema:* we append the fixed rubric R (consensus → trust margin → close clusters → trust-sum, with clinical tie-break) and instruct the LLM to return *JSON only* with keys {`final_class`, `reason`}. We enforce a strict schema with `final_class`∈ {N0, N+(1–2), N+(>2)}; any extra prose or malformed JSON triggers a re-prompt. For readability, we show in Fig. 4 a simplified overview of the prompt structure, summarizing the expert table, optional clinical block, and the arbiter's rubric and JSON schema in a left-to-right flow.

Let C denote the class set, with the LLM receives, in text, the tuples $\{(\hat{y}_j, c_j, \tau_j)\}_{j=1}^{M}$, optional clinical summary s_{clin}, and a rubric $\mathcal{R}$ describing preferences over these quantities. The LLM outputs

$$\text{LLM}\big(\{\hat{y}_j, c_j, \tau_j\}_{j=1}^{M}, s_{\text{clin}}, \mathcal{R}\big) \rightarrow \big(\hat{y}_{\text{final}}, \text{reason}\big),$$

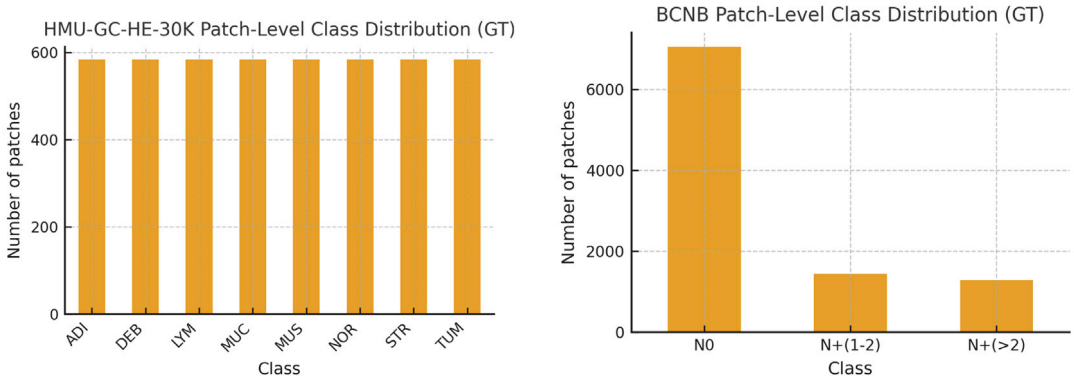
where $\hat{y}_{\text{final}} \in C$ and reason is a short natural-language explanation grounded in experts' signals and (if present) clinical evidence. In practice, we enforce a strict output schema (JSON keys and allowed class strings) to ensure reliable parsing, while keeping the reasoning free-form enough to be human-readable.

In MELLMA, arbitration occurs only at the patch level. Each patch extracted from a whole-slide image is independently analyzed by the M vision agents, followed by trust estimation and LLM arbitration to obtain a final class and rationale. Patient-level decisions are then derived by aggregating all patch-level outputs using two complementary rules: (i) majority vote of the patch-level final classes; and (ii) trust-weighted probability pooling,

$$P_{\text{patient}}(y = k) = \frac{1}{|P|} \sum_{i \in P} \sum_{j=1}^{M} \tau_{ij} \, p_{ij}(y = k), \tag{3}$$

where P indexes all patches belonging to that patient, $p_{ij}(y = k)$ is the predicted probability from agent j for patch i, and τ_{ij} is its corresponding trust score. The patient-level label is then assigned as:

$$\hat{y}_{\text{patient}} = \arg\max_k \; P_{\text{patient}}(y = k).$$



(a) (b)

Fig. 5. Class distribution. In (a) HMU-GC-HE-30K, and (b) BCNB datasets. The dataset HMU-GC-HE-30K comprises of ∼31,000 H&E-stained gastric cancer tissue patches annotated into eight tumor microenvironment classes equally distributed — adipose tissue (ADI), debris (DEB), mucus (MUC), muscle (MUS), lymphocyte aggregates (LYM), stroma (STR), normal mucosa (NOR), and tumor epithelium (TUM). The dataset BCNB includes ∼41,000 patches obtained from core-needle biopsy WSIs with corresponding clinical metadata. Patches are annotated into three nodal categories (N0, N+(1–2), N+(>2)), showing a strong imbalance where N0 dominates. For binary classification, N+(1–2) and N+(>2) are grouped as N+. This imbalance motivates both multi-class and binary evaluations in our experiments.

For binary BCNB evaluation (N+ vs. N0), we threshold the proportion of patch-level N+ predictions to obtain the final patient-level label. No additional LLM inference is performed at the patient level; aggregation is deterministic and computationally efficient. The LLM is carefully applied only at stages where expert disagreement is most pronounced, enabling contextual arbitration among conflicting predictions. For the final diagnostic decision, we adopt a transparent aggregation rule to avoid introducing additional sources of variability and to preserve interpretability. This design isolates the contribution of LLM-guided arbitration and prevents error propagation that could arise from cascading multiple LLM decision stages.

Whole-slide images often contain only a small number of abnormal patches while the majority correspond to normal tissue. To reduce dilution effects during patient-level aggregation, MELLMA does not rely solely on majority voting. The trust-weighted pooling rule (Eq. (3)) assigns higher influence to patches with strong expert agreement and high meta-learner trust, thereby amplifying diagnostically informative regions. For the binary BCNB setting, we replace majority voting with a proportion-based threshold on the number of N^+ patches, which prevents abundant normal tissue from suppressing focal metastatic regions. This hierarchical design improves robustness when abnormalities are spatially localized.

3.5. Datasets and modality usage

We evaluate MELLMA on two complementary histopathology datasets with distinct modalities and class structures. HMU–GC–HE–30K contains 300 WSIs from 300 patients, producing approximately 31,000 non-overlapping 224 × 224 H&E patches annotated into eight tumor microenvironment classes. BCNB consists of 1058 patients with around 41,000 tumor-region patches (512 × 512), annotated into three nodal categories (N0, N+(1–2), N+(>2)) and also used in a binary N0 vs. N+ setting. All experiments use strict patient-level train/validation/test splits to prevent any patch-to-patch leakage across subsets with 70/10/20 for HMU–GC–HE–30K and 80/10/10 for BCNB. Class distributions and patch counts for both datasets are illustrated in Fig. 5, and the same data partitions are used consistently across all expert models, meta-learner variants, and LLM arbiters.

1. HMU-GC-HE-30K (Lou et al., 2025): The HMU-GC-HE-30K dataset is a gastric cancer histopathology collection containing 300 WSIs from 300 patients, which are tiled into approximately 31,000 patches of size 224 × 224 pixels. Each patch is annotated into one of eight classes of tissues in the tumor microenvironment (TME), namely, adipose tissue (ADI), debris (DEB), mucus (MUC), muscle (MUS), lymphocyte aggregates (LYM), stroma (STR), normal mucosa (NOR), and tumor epithelium (TUM), by expert pathologists. The dataset captures a wide variety of gastric tissue histology under H&E staining with slide sizes that vary widely. The dataset includes slide level clinical metadata alongside patches, but the published resources do not clearly report patch-to-slide ID mapping for each tile; thus, we focus on patch level analyses and the models trained on purely image evidence.
2. BCNB (Xu et al., 2021): The early breast cancer core-needle biopsy (BCNB) dataset includes whole-slide core needle biopsy WSIs from 1058 patients (41,000 patches) each with corresponding clinical data. Clinical metadata per patient include age, tumor size, histological and molecular subtypes (ER, PR, HER2), Ki67 index, number of lymph node metastases, and ALN (axillary lymph node) status. In total three classes (N0, N+(1-2), N+(>2)) were covered. Tumor regions in WSIs are annotated by two pathologists, and cores are tiled or patched for model input (patch size 512 × 512), with patches filtered for blank/background content. The dataset shows class imbalance, especially in high-metastasis categories, making performance on minority classes particularly important. Magnification or scanning resolution is consistent across samples, and the clinical data is desensitized and non-private. Class distribution is imbalanced; for example, the N0 category contains the majority of patches compared with the other classes, creating challenges for minority-class discrimination. To address this, in addition to the three-class formulation, we also considered a binary classification setting where N+(1–2) and N+(>2) were grouped as N+, which helps to mitigate the high class imbalance.

In the proposed architecture we fuse complementary experts to inject a learned reliability signal, and constrain LLM reasoning under a transparent rubric. Our design includes (i) complementary visual expertise from multiple agents, (ii) a data-driven, per-case trust estimator, and (iii) the LLM's structured reasoning. The trust-weighted mixture P_{ens} (Eq. (2)) provides a stable base distribution; the LLM then reconciles residual conflicts under a fixed rubric, staying aligned with quantitative evidence and, on BCNB, the clinical summary s_{clin} when ties arise. The result is a robust, auditable fusion in which decisions reflect quantitative reliability (agreement, confidence, trust)

and clinical plausibility, and each case is accompanied by a concise natural language rationale. As we show in our ablation experiments (Section 4.5), removing the trust signal degrades accuracy and shifts explanations toward simple consensus, supporting this design choice.

4. Experiments

4.1. Experimental setup

We conducted a comprehensive evaluation of the proposed multi-agent LLM-arbitrated classifier using the HMU–GC–HE–30K and BCNB datasets. Following standard practice in computational pathology, the datasets were divided into training, validation, and test sets at the patient level to avoid data leakage across patches from the same individual. For training the base agents, we used 224 × 224 pixel patches for the HMU-GC-HE-30K and 512 × 512 pixel patches for the BCNB extracted from whole slide images (WSIs), with non-informative regions filtered out and stain normalization applied to mitigate inter-laboratory color variation. Four agents representing diverse architectures were selected - ResNet-50, ViT-B/16, EfficientNet-B3, and Swin-V2-S. These models were chosen to span complementary inductive biases (local textures vs. global context) to match common pathology baselines with stable pretrained weights, and fit in our throughput experiment. Additionally, preliminary runs with heavier models exhibited similar trends, so we focus on this representative set.

Inference was performed on the test set patches, based on per-agent predictions, confidence values, and trust scores derived from the meta-learner. The meta-learner was trained on the predictions generated by the expert models on the training set and evaluated on the validation set to prevent data leakage. These trust scores quantify the reliability of each agent for a given patch and are critical for arbitration. For computational infrastructure, all experiments were run on a Linux server equipped with NVIDIA V100 GPUs with 128 GB RAM, PyTorch 2.1, Hugging Face Transformers 4.41, and typically CNN models took on average 2 h, and ViT models took on average 3 h for a maximum of 100 epochs. The large language models (LLMs) used for arbitration were accessed as follows: LLaMA-3 8B (local deployment), Mistral-7B Instruct (local checkpoint), GPT-4o (via OpenAI API) and GPT-4 (via OpenAI API). We selected these four arbitrators to span both open-source, locally deployable models and frontier API models, enabling a controlled comparison across deployment constraints and reasoning capacity. Here, LLaMA-3 (8B) and Mistral-7B represent light-weight, reproducible options. GPT-4 and GPT-4o represent state-of-the-art proprietary models with stronger instruction follow-up and calibration. The arbitration step was implemented in Python and integrated seamlessly with agent outputs and clinical metadata.

For completeness, we provide the training configuration used for all expert models. Each model was initialized with ImageNet-1K pretrained weights and trained using AdamW (weight decay 1×10^{-4}) with a cosine learning-rate schedule and warmup. We used a batch size of 32 and monitored validation performance every epoch, applying early stopping after 8 epochs without improvement. Data augmentation included random horizontal and vertical flips, small rotations, color jitter, and mild stain perturbations. Patient-level splits were strictly enforced so that no patches from a given patient appeared across the training, validation, and test sets. Patient-level outcomes are obtained by deterministic aggregation of patch-level LLM predictions (see Eq. (3) in Section 3.4). We evaluated performance at both the patch level and the patient level, the latter being most clinically relevant. Patient-level predictions were obtained by majority voting and trust-weighted aggregation of patch-level outputs per case. We first run patch-level arbitration in which LLM produces a final class per patch. Patient level predictions are then obtained by majority voting over patch-level final classes; in the binary setting, we threshold the fraction of patch-level N+ predictions (N+(1–2) or N+(>2)) to produce the patient label. Note that there are no patient-level LLM calls. For evaluation metrics, we followed best practices in the computational pathology domain and report accuracy, area under the receiver operating characteristic curve (ROC–AUC) score, Binary accuracy, and AUC scores and weighted F1 score, covering both multi-class and binary formulations. For example, for the BCNB dataset we also considered N0 versus any N+ classes. This diverse set of metrics provides a rigorous assessment of class imbalance, agreement, and discriminative ability.

4.2. Comparative baselines

To establish a solid baseline, we first evaluated the four individual agents for the BCNB (ResNet-50, ViT-B/16, EfficientNet-B3, and Swin-V2-S) and three for the HMU-GC-HE-30K (ResNet-50, ViT-B/16, and EfficientNet-B3). Finally, we evaluated LLM based arbitration strategies, in which LLMs were assigned with reading agent predictions, trust scores, and available clinical metadata to output a final class with a brief textual justification. The three arbitrators were LLaMA-3, GPT-4o, GPT-4 and Mistral-7B. Their outputs were parsed into structured predictions and recorded along with their free text reasoning. Importantly, no deterministic rules (e.g., if-else logic) were used, and the LLM was solely responsible for combining evidence, making arbitration decisions, and generating interpretable reasons. We report single agent models (ResNet-50, EfficientNet-B3, ViT-B/16, SwinV2-S), ensembling methods (majority vote, uniform average), and our LLM-arbitrated pipeline. We further consider two internal trust-only variants (1) meta-pick (choose the highest trust agent), and (2) meta-ensemble (trust-weighted probability mixture), as diagnostics for the meta-learner. In addition to single-expert baselines, we include two trust-only fusions, namely the meta-pick and meta-ensemble to quantify what a learned trust signal achieves without any LLM reasoning. Both were consistently below the LLM arbiters across metrics and we use them only as sanity checks. In HMU–GC–HE–30K, we therefore report three expert baselines (ResNet–50, ViT–B/16, EffNet–B3), while the fourth backbone SwinV2-S is only used in the BCNB experiments (Tables 2 and 3) to keep the HMU experiments computationally tractable and focused on the most representative CNN/Transformer backbones. First, HMU consists of more than 31,000 patches, and the full pipeline (agents → trust estimator → LLM arbitration) incurs substantial computational overhead; adding a fourth expert would proportionally increase inference cost without providing clear architectural benefit. Second, SwinV2-S did not provide performance advantages in our BCNB experiments, and therefore offered limited additional value for the more homogeneous, image-only HMU setting. Thus, the three-expert configuration represents a balanced and computationally efficient setup while preserving architectural diversity.

4.3. Quantitative results

HMU-GC-HE-30K (patch-level, multi-class). Table 1 shows that LLM arbitration improves substantially over individual agents in both accuracy and F1 metrics. The LLaMA-3 arbitrator achieves the best overall accuracy of 0.796 and Weighted-F1 of 0.795, while outperforming the strongest single agent (ResNet-50, 0.732 Acc) by approximately 8.7% relative F1 (0.781) and AUC scores (0.783) also indicate better balanced performance across classes than any agent. In terms of accuracy, GPT-4 (0.778) and GPT-4o (0.769) are competitive, with Mistral-7B close behind (0.770). In particular, agents' AUCs peak around 0.74–0.75, whereas LLM arbitrators cluster approximately at 0.78, reflecting more consistent ranking of true class over confounders. The takeaway is that even without any clinical metadata, LLM-guided arbitration provides a robust, calibrated improvement over classic CNN/Transformer experts. As comparative baselines, the meta-pick and meta-ensemble along with the majority vote and uniform average outperform individual agents, however these methods remain below the LLM arbitrators, indicating that our proposed MELLMA framework adds value beyond static or trust-only fusion.

Table 1
Patch-level performance on the HMU-GC-HE-30K dataset. We report Accuracy, Weighted-F1, Macro-F1, and AUC. Best results in are in bold.

Method	Accuracy	Weighted-F1	Macro-F1	AUC
ResNet50	0.732	0.731	0.715	0.734
ViT-B16	0.727	0.728	0.712	0.739
EffNet-B3	0.713	0.714	0.697	0.746
Majority Vote	0.757	0.758	0.758	0.748
Uniform Avg (prob)	0.758	0.759	0.759	0.748
Meta-Pick (trust top-1)	0.758	0.758	0.758	0.716
Meta-Ensemble (trust-weighted)	0.760	0.761	0.761	0.748
LLM arbitrator (LLaMA-3)	**0.796**	**0.795**	**0.781**	**0.783**
LLM arbitrator (GPT-4)	**0.778**	**0.764**	**0.771**	**0.780**
LLM arbitrator (GPT-4o)	**0.769**	**0.770**	**0.770**	**0.781**
LLM arbitrator (Mistral)	**0.770**	**0.775**	**0.775**	**0.778**

Table 2
BCNB: Patch level multi-class (3 classes: N0, N+(1–2), N+(>2)) and binary (N+ vs N0) results. We report Multi-class Accuracy, AUC, Weighted-F1; and Binary Accuracy, AUC for individual agents (ResNet50, ViT-B/16, EfficientNet-B3, Swin-V2-S) and LLM arbitrators (LLaMA-3, GPT-4o, GPT-4, Mistral). Rows for Majority Vote, Uniform Avg, Meta-Pick, and Meta-Ensemble are LLM-independent and computed once from agent outputs and meta-trust. Best results are in bold (per column).

Method	MC Acc	OvR AUC	Weighted F1	Bin Acc	Bin AUC
ResNet50	0.677	0.681	0.691	0.745	0.719
ViT-B16	0.655	0.632	0.665	0.684	0.628
EffNet-B3	0.675	0.676	0.687	0.736	0.728
Swin-V2-S	0.543	0.642	0.609	0.446	0.746
Majority Vote	0.694	0.658	0.660	0.748	0.727
Uniform Avg (prob)	0.695	0.700	0.661	0.747	0.731
Meta-Pick (trust top-1)	0.727	0.690	0.663	0.754	0.715
Meta-Ensemble (trust-weighted)	0.711	0.652	0.665	0.752	0.698
LLM arbitrator (LLaMA-3)	**0.748**	**0.717**	0.721	**0.798**	**0.748**
LLM arbitrator (GPT-4o)	0.727	0.706	**0.732**	0.765	0.727
LLM arbitrator (GPT-4)	0.730	0.701	0.728	0.771	0.732
LLM arbitrator (Mistral)	0.725	0.706	0.702	0.792	0.733

Table 3
BCNB: Patient level multi-class (3 classes: N0, N+(1–2), N+(>2)) and binary (N+ vs N0) results. We report Multi-class Accuracy, AUC, Weighted-F1; and Binary Accuracy, AUC for individual agents (ResNet-50, ViT-B/16, EfficientNet-B3, Swin-V2-S), multiple instance learning (MIL) approaches such as ABMIL, DSMIL, TransMIL, and LLM arbitrators (LLaMA-3, GPT-4o, GPT-4, Mistral). Rows for Majority Vote, Uniform Avg, Meta-Pick, and Meta-Ensemble are LLM-independent and computed once from agent outputs and meta-trust. Best per column is bold.

Method	MC Acc	OvR AUC	Weighted F1	Bin Acc	Bin AUC
ResNet-50	0.646	0.713	0.667	0.703	0.742
ViT-B16	0.638	0.705	0.686	0.677	0.698
EffNet-B3	0.677	0.708	0.705	0.668	0.715
Swin-V2-S	0.512	0.644	0.519	0.425	0.720
Majority Vote	0.630	0.712	0.587	0.677	0.755
Uniform Avg (prob)	0.653	0.741	0.609	0.701	0.784
Meta-Pick (trust top-1)	0.677	0.741	0.600	0.693	0.754
Meta-Ensemble (trust-weighted)	0.677	0.744	0.623	0.701	0.780
ABMIL	0.685	0.701	0.531	0.697	0.711
DSMIL	0.704	0.730	0.589	0.719	0.741
TransMIL	0.711	0.736	0.592	0.723	0.758
LLM arbitrator (LLaMA-3)	0.696	**0.747**	0.716	**0.745**	**0.803**
LLM arbitrator (GPT-4o)	**0.715**	0.738	0.711	0.724	0.781
LLM arbitrator (GPT-4)	0.712	0.735	0.709	0.726	0.785
LLM arbitrator (Mistral)	0.701	0.728	**0.735**	0.721	0.779

BCNB (patch-level, multi-class and binary). At the BCNB patch-level, arbitration markedly improves over individual agents as shown Table 2. LLaMA-3 delivers the strongest multi-class accuracy (0.748) and a competitive Weighted-F1 (0.721), exceeding the best agent baseline (ResNet-50, 0.677 Acc) by approximately 7.4%. GPT-4o (0.727 Acc, 0.732 W-F1) and GPT-4 (0.730 Acc, 0.728 W-F1) also achieve substantial gains, confirming the robustness of arbitration across different LLMs. For AUC, Mistral-7B yields the highest value (0.706), suggesting stronger balance across minority classes, although its accuracy (0.725) lags slightly behind. In the binary N+ detection setting, all LLM arbitrators surpass the agents in headline metrics. Mistral-7B attains the highest AUC (0.733), while LLaMA-3 achieves the best binary accuracy (0.798). LLaMA-3 provides the strongest joint performance (0.798 Acc, 0.748 AUC), indicating that its arbitration choices are both consistent and clinically relevant. GPT-4o offers a balanced trade-off (0.765 Acc, 0.727 AUC). Overall, arbitration consistently strengthens both multi-class recognition and binary stratification, with different LLMs showing complementary strengths across accuracy, F1, and AUC. The meta only baselines such as the meta-pick, and meta-ensemble obtain values between single agents and LLM arbitrators, confirming that learned trust helps without LLM reasoning, however the full LLM arbitration provides the largest gains.

BCNB (patient-level, multi-class and binary). At the patient level, aggregating patch evidence before arbitration amplifies improvements as detailed in Table 3. GPT-4o achieves the highest multi-class accuracy (0.715) with Weighted-F1 of 0.711, exceeding the best agent baseline (EffNet-B3, 0.677 Acc) by approximately 4%. LLaMA-3 delivers the strongest Weighted-F1 (0.716) and the highest binary AUC (0.803), underscoring its robustness when predictions are aggregated. GPT-4 performs consistently (0.712 Acc, 0.709 W-F1, 0.785 Bin AUC), while Mistral-7B achieves the best macro AUC (0.763) and a solid Weighted-F1 (0.735). For the binary N+ task, patient level arbitration produces clinically meaningful discrimination. LLaMA-3 leads in binary accuracy (0.745) and AUC (0.803), while GPT-4o (0.724 Acc, 0.781 AUC) and GPT-4 (0.726 Acc, 0.785 AUC) remain strong contenders. Mistral-7B is competitive as well (0.721 Acc, 0.779 AUC). Taken together, these results demonstrate that arbitration at the patient level not only improves overall classification accuracy but also enhances the ability to reliably separate N+ from N0/N–, reinforcing its practical relevance in clinical decision support. We note that although Swin-V2-S is the weakest individual expert, its impact on arbitration is minimal because the meta-learner consistently assigns lower trust to this model on ambiguous patches. Removing Swin-V2-S from the expert pool produced only negligible changes in both patch- and patient-level arbitration accuracy, indicating that the arbitration mechanism is robust to weaker experts. Multiple instance learning (MIL) is a widely practiced algorithmic family for patient-level diagnosis in computational pathology and addresses how instance-level features are aggregated into a bag representation during model training. To provide a fair reference, we additionally include representative MIL models namely ABMIL (Ilse et al., 2018), DSMIL (Li et al., 2021), and TransMIL (Shao et al., 2021) in Table 3, as commonly used reference methods, trained and evaluated using the same data split as the rest of this work (80%/10%/10% on BCNB) under the same experimental protocol as the other methods. Importantly, our goal is not to replace MIL, but to study a complementary problem, how to arbitrate disagreements among heterogeneous classifiers that already produce task-specific predictions. While MIL focuses on learning an implicit aggregation function, MELLMA operates at the decision level, explicitly reconciling model outputs using learned trust signals and structured LLM reasoning. Similar to the trends observed in the patch level, the meta only baselines again obtained intermediate values, whereas the LLM arbitrators deliver the strongest performance over all metrics after aggregating patch evidence. We note that these results represent a consistent performance of our proposed MELLMA approach across patch and patient-level.

4.4. Qualitative analysis of arbitration reasoning

Unlike static ensembles, which only output probabilities, LLMs articulate explicit rationales when reconciling agent disagreements. Table 4 and Table 5 summarize representative arbitration cases for the BCNB (multi-modal) and HMU-GC-HE-30K (image-only) datasets, respectively. Each row compares how LLaMA-3, GPT-4o, GPT-4, and Mistral-7B justified their final prediction when faced with conflicting agent output.

Table 4 shows that on the BCNB dataset, GPT-4o consistently produced structured, clinically aligned explanations. Its reasonings frequently reference both majority agreement and patient metadata (e.g., ER/PR/HER2 status, Ki-67 index), resulting in decisions that are both accurate and clinically plausible. GPT-4 showed similar behavior in many cases, though its reasoning was occasionally less calibrated; in some examples, it over-weighted clinical context and diverged from the agent consensus, leading to weaker predictions, for example the last two rows. LLaMA-3 remained competitive, typically emphasizing consensus among trusted agents and arriving at correct outcomes in most cases. Mistral-7B, in contrast, often defaulted to a single agent's prediction without deeper justification, highlighting its limitations in reasoning depth.

From Table 5 we note that on HMU-GC-HE-30K wherein no clinical metadata was available, the differences in reasoning style remained evident. GPT-4o again provided the most balanced reasoning, carefully weighing agent agreement and trust margins. GPT-4 and LLaMA-3 were competitive, though GPT-4 occasionally over-interpreted sparse signals, and LLaMA-3 tended to give shorter, consensus-driven rationales. Mistral-7B frequently defaulted to whichever agent showed slightly higher trust, even when this contradicted the majority. Importantly, across both datasets, GPT-4o and LLaMA-3 demonstrated greater robustness in ambiguous cases, aligning their reasoning with both quantitative evidence and plausible clinical logic.

4.5. Ablation studies

We studied the meta-learner's contribution on BCNB by running arbitration with identical inputs and prompts while removing the trust vector from the LLM context. We restrict ablations to BCNB because HMU-GC-HE-30K lacks patch to slide identifier and relevant clinical metadata, which prevents patient-level and clinical cue ablations. This ablation examines whether learned per-agent reliability and corresponding clinical cues actually change both performance and the policy of the arbitrators.

Disabling trust degrades both patch and patient-level performances of the LLMs as shown in Table 6. For example, LLaMA-3 drops from 0.748 to 0.687 (patch accuracy) and from 0.696 to 0.605 (patient accuracy); its patch AUC and patient AUC also decline. GPT-4 and GPT-4o show the same trend. For example, patch accuracy from 0.730 to 0.496 for GPT-4 and from 0.727 to 0.510 for GPT-4o. Mistral-7B shows the largest relative loss (0.725 to 0.591). At patient level, the accuracy scores goes down from 0.696 to 0.605 for LLaMA3, 0.712 to 0.681 for GPT-4, 0.715 to 0.681 and 0.701 to 0.570 for Mistral-7b. Similar trend can be seen for the AUC scores for both pathc and patient level. This consistent decrease indicate that learned trust not only increases confidences but also it enables the LLM to override misleading majorities in hard cases.

As can be seen in Table 7 without the meta learner, trust-centric reasoning disappears and explanations become more toward consensus/confidence. In practice, the proportion of reasons mentioning "trust" falls to 0% for every LLM, while mentions of consensus/confidence rise noticeably. Decision stage self reports also change. Models "exit" earlier under Unanimous/majority conditions and invoke fewer tie-break operations.

Using the same BCNB patches as shown Table 4, Table 8 shows how reasons change when trust scores are OFF. Under trust scores ON, arbiters often cite a dominant trust margin (plus clinical alignment) to select the correct minority expert in ambiguous 2–2 or 2–1–1 splits. With trust scores OFF, the same cases are resolved via majority or confidence alone, and the reasons no longer reference trust margins. This paired view confirms that the performance deltas in Table 6 are causally linked to a shift in arbitration policy rather than superficial wording.

Although the BCNB dataset provides patient-level clinical descriptors, these variables are low-dimensional and relatively coarse (e.g., ER/PR/HER2 status, Ki67 index, tumor size). In our experiments we observed that the LLM arbitrator remains fully functional even when the clinical block is removed from both the meta-learner input and the LLM prompt. The corresponding performance differences were marginal, which is consistent with prior studies on BCNB where biopsy morphology is the dominant predictive factor. Because the HMU–GC–HE–30K dataset does not contain clinical metadata, multimodal ablation is inherently restricted to BCNB. These observations indicate that the modest multimodal gains are driven primarily by dataset limitations rather than by constraints of the proposed framework, and that the arbitration mechanism generalizes naturally to both image-only and image+clinical settings.

Table 4
Qualitative analysis of LLM arbitration reasoning (BCNB). All rows are the same cases across models.

ID	GT	Agent predictions (summary)	LLaMA-3 reason	GPT-4o reason	GPT-4 reason	Mistral-7B reason
Patch_1123	N+(1–2)	ResNet=N+(1–2), ViT=N+(1–2), EffNet=N0 (two agree, high trust)	Two agents agree with high confidence; choose N+.	Consensus of ResNet & ViT; metadata consistent with N+. Final: N+(1–2).	Despite the clinical context of ER-Negative and PR-Positive, the majority of agents predicted N+(1–2) with a higher total trust.	Picked the highest-trust agent (ResNet).
Patch_2487	N0	ResNet=N0, ViT=N0, EffNet=N+	Majority vote indicates N0.	ResNet & ViT predict N0 with strong trust; metadata aligns. Final: N0.	The majority of agents predicted N0 with high trust and confidence, and the clinical context (ER=Positive; PR=Positive; HER2=Negative; Age=68; Ki67=<5%) aligns with lower nodal burden.	Followed ResNet due to higher trust value.
Patch_3375	N+(>2)	ResNet=N+(>2), ViT=N0, EffNet=N+(>2)(low trust spread)	Two agents lean N+(>2); assign N+(>2).	ResNet & EffNet predict N+(>2); trust spread acceptable. Final: N+(>2).	All agents unanimously predicted N+ with high confidence and trust, except ViT and the clinical context (ER=Positive; PR=Positive; HER2=Negative; Age=82; Ki67=0.2) supports a lower nodal burden.	Selected N0 because ViT had slightly higher trust.
Patch_4199	N0	Mixed: ResNet=N+, ViT=N+, EffNet=N0	Leaning to EffNet's N0 due to trust margin.	Despite two N+ votes, EffNet's calibrated trust & metadata favor N0. Final: N0.	The class N0 has the highest total trust and agent consensus, and the clinical context of ER and PR positive with low Ki67 supports lower nodal burden.	Followed ViT's N+ without justification.
Patch_5021	N+(1–2)	ResNet=N+, ViT=N0, EffNet=N0	Difficult case; favor N0.	Trust spread shows ResNet slightly higher; metadata aligns with N+. Final: N+(1–2).	Agents confidently predicted N0, supported by negative ER, PR, HER2 and low Ki67 indicating lower nodal burden.	Chose N0 because majority agents said N0.
Patch_6244	N0	ResNet=N0, ViT=N+, EffNet=N+	Conflicting votes; confidence for N+ not strong.	Despite two N+ votes, ResNet trust & context indicate N0. Final: N0.	Majority of agents predicted N+(>2) with higher total trust, and the clinical context of ER+, PR+, HER2+ and high Ki67 suggests a higher nodal burden.	Majority says N+, so pick N+.

Table 5
Qualitative analysis of LLM arbitration reasoning on HMU-GC-HE-30K. Cases highlight differences in reasoning style and prediction quality.

Example ID	GT	Agent predictions (summary)	LLaMA-3 reason	GPT-4o reason	GPT-4 reason	Mistral-7B reason
Patch_1402	A	ResNet=A, ViT=A, EffNet=B	Two agents agree, trust supports A → Final A.	Consensus of ResNet & ViT, both high confidence → A.	Strong majority for A; margin over EffNet is clear → A.	Picked ResNet's prediction (A).
Patch_2215	C	ResNet=C, ViT=B, EffNet=C	Two agents lean C, acceptable spread → Final C.	ResNet & EffNet agree on C; trust favors C → Final C.	Tie-breaker via confidence margin → C.	Chose B because ViT had slightly higher trust.
Patch_3159	B	ResNet=A, ViT=B, EffNet=B	Majority = B → Final B.	ViT & EffNet both vote B, consistent trust → B.	B is reinforced by ensemble trust → Final B.	Selected A due to ResNet alone.
Patch_4266	D	ResNet=D, ViT=C, EffNet=D	Two votes for D, so D.	Agreement between ResNet & EffNet, higher trust; confident D.	ResNet+EffNet outweigh ViT → Final D.	Followed ViT's C.
Patch_5180	A	Mixed predictions, no consensus	Hard case; picked B.	Though agents split, ResNet trust margin favors A → Final A.	Calibrated trust suggests A more likely than C → A.	Randomly chose C.
Patch_6024	C	ResNet=C, ViT=A, EffNet=C	Two agents agree, class C.	ResNet+EffNet consensus and trust spread → C.	Notes consistent signals for C → Final C.	Picked A without justification.

5. Discussion

Across HMU-GC-HE-30K and BCNB, LLM arbitration improves both performance and explanations. On HMU-GC-HE-30K we see 7%–9% accuracy gains over the best agent, and on BCNB the arbitrators reach 0.748 patch accuracy and 0.747 patient level AUC. Ablations confirm that the meta-learner's trust is the driver where Trust-ON >Trust-OFF for all the LLM models (e.g., for LLaMA-3 from 0.687 to 0.748 patch acc), with reasons shifting from consensus/confidence to trust-guided. Our results across two datasets underscore three main observations.

- LLM arbitration consistently improves over individual agents. In HMU-GC-HE-30K, even without clinical metadata, arbitrators such as LLaMA-3 and GPT-4o increased the accuracy and weighted F1 by 7%–9% relative to the best individual agent. This

Table 6
BCNB - LLM arbitration (multi-class). Metrics at patch- and patient-level; Trust-OFF (no meta-learner trust).

Method	Patch Acc	Patch AUC	Patient Acc	Patient AUC
LLaMA-3				
LLM arbitrator	**0.687**	**0.5**	**0.605**	**0.519**
Trust-OFF (LLaMA-3)				
LLM arbitrator	0.748	0.717	0.696	0.747
Trust-ON (LLaMA-3)				
GPT-4				
LLM arbitrator	**0.496**	**0.668**	**0.681**	**0.724**
Trust-OFF (GPT-4)				
LLM arbitrator	0.730	0.701	0.712	0.735
Trust-ON (GPT-4)				
GPT-4o				
LLM arbitrator	**0.510**	**0.668**	**0.681**	**0.726**
Trust-OFF (GPT-4o)				
LLM arbitrator	0.727	0.697	0.715	0.738
Trust-ON (GPT-4o)				
Mistral-7b				
LLM arbitrator	**0.591**	**0.662**	**0.570**	**0.731**
Trust-OFF (Mistral-7b)				
LLM arbitrator	0.725	0.706	0.701	0.728
Trust-ON (Mistral-7b)				

Table 7
BCNB - Explanation shift when the meta-learner is disabled (Trust OFF). Percentages are computed over all arbitrated patches. "Decision step": Unanimous (4–0); 3–1 majority; 2–2/2–1–1 split (tie broken via confidence/clinical cues); Fallback (no majority, heuristic); Tie (unresolved).

[l]LLM (Trust OFF)	Arbitrated patches	% Trust	% Consensus	% Confidence	% Clinical	Median words	Decision step (%)				
							Unanimous	3–1	2–2/ 2–1–1	Fallback	Tie
GPT-4	3500	0.0	30.4	90.8	88.1	17	75.1	12.3	3.4	9.1	0.0
GPT-4o	3500	0.0	52.4	95.2	99.1	17	67.8	18.4	11.7	0.0	2.1
LLaMA-3	3500	0.0	93.4	91.6	65.2	11	95.4	1.6	0.0	0.2	2.7
Mistral-7B	3500	0.0	0.0	71.2	100.0	11	36.2	42.4	21.4	0.0	0.0

suggests that arbitration is not merely averaging predictions but leveraging nuanced inter agent agreement patterns to produce more robust outcomes.

- The multi-modal arbitration brings clinically meaningful advantages. On the BCNB dataset, where patient metadata was included, arbitration achieved gains not only in patch level performance but also in patient level aggregation. LLaMA-3 attained the best binary AUC (0.803), indicating strong discrimination of nodal metastasis status, while GPT-4o provided the best patient-level accuracy (0.715). These results align with clinical expectations: when multiple weak signals (patch evidence) are combined with structured metadata (e.g., ER, PR, HER2), the resulting inference becomes more reliable and better reflects real diagnostic reasoning.
- Different LLMs perform well in different ways. GPT-4o often produced the most balanced trade-off between accuracy and AUC, LLaMA-3 favored higher sensitivity and accuracy in certain settings, and Mistral-7B, while trailing slightly, offered stronger AUC values in minority class discrimination. This diversity highlights that arbitration outcomes are model dependent and suggests that multiple arbiters could be deployed in parallel for ensemble of arbitrators strategies in future work.

Beyond architectural considerations and lesion localization challenges, it is also important to clarify how the proposed framework relates to existing learning strategies such as multiple instance learning (MIL). We emphasize that MELLMA is not intended as a replacement for MIL-based pipelines. MIL addresses how instance-level evidence is aggregated during representation learning, whereas MELLMA focuses on decision-time arbitration among heterogeneous predictors using trust estimation and LLM-guided reasoning. These two approaches are orthogonal: MIL models can naturally serve as experts within the proposed arbitration framework, and future work may explore LLM-guided arbitration over mixtures of MIL and non-MIL models.

In practical pathology workflows, malignant or abnormal areas may occupy only a very small fraction of the slide. Because MELLMA performs reliable patch-level arbitration using multiple experts and trust signals, only a few abnormal patches are needed to trigger an accurate case-level prediction. The trust-weighted aggregation further mitigates dilution by down-weighting low-trust normal regions. For BCNB, the proportion-based N^+ threshold ensures that localized metastatic regions are not overlooked due to the predominance of normal tissue. This strategy aligns with clinical expectations, where even small metastatic foci are diagnostically meaningful.

Based on our qualitative results, we can highlight two key points. First, arbitration is not the only numerically superior, it is also more interpretable. The stronger LLMs explain why they favored particular agents, improving transparency and clinical trust. Second, the variability across LLMs illustrates that reasoning depth matters. Higher capacity models (GPT-4, GPT-4o) generate explanations closer to human clinical reasoning, while lighter models (Mistral-7B) lack the same interpretive nuance. Together, these findings reinforce that LLM arbitration advances both performance and explainability, addressing two critical requirements for adoption in computational pathology. Tables 4 and 5 demonstrate that LLMs not only decide final classes but also articulate concise, human readable reasonings. GPT-4o explanations over BCNB dataset frequently incorporated both agent consensus and clinical cues, producing justifications that align closely with how oncologists weigh evidence (e.g., majority agent vote reinforced by ER/PR/HER2 status). LLaMA-3 also provided competitive reasoning, emphasizing inter agent trust patterns. In contrast, Mistral-7B tended to over rely on single agent trust margins, producing weaker explanations and, in some cases, incorrect predictions. GPT-4, while strong in several cases, sometimes overweighted clinical features relative to imaging, occasionally diverging from ground truth. For the HMU-GC-HE-30K dataset, where no clinical metadata was present, GPT-4o again displayed the

Table 8
BCNB - Same patches as Table 4, but with the meta-learner disabled (Trust OFF). For each patch and LLM, we show the predicted class (Pred), the model's short explanation parsed from the CSV, and the self-reported decision stage (*Step*): *Unanimous* (4–0 agreement), *3–1 majority*, *2–2/2–1–1 split* (tie broken via confidence/clinical cues), *Fallback* (no majority; heuristic), *Tie* (still unresolved).

Patch	GT	LLM	Pred	Step	Reason (Trust-OFF)
1123	N0	GPT-4	N+(>2)	Unanimous	Majority of agents predicted N+(>2) with high confidence, and clinical context supports higher nodal burden.
1123	N0	GPT-4o	N+(>2)	3–1 majority	Agent consensus favors N+(>2) with high confidence from ResNet50... clinical context (HER2+ and high Ki67) supports higher nodal burden.
1123	N0	LLaMA-3	N+(>2)	Unanimous	ResNet50 and EffNetB3 agree on N+(>2) with high confidence, supported by clinical context.
1123	N0	Mistral-7B	N+(>2)	3–1 majority	3/4 agents (ResNet50, EffNetB3, Swin) vote N+(>2) with higher confidence; clinical cues indicate higher nodal burden.
2487	N0	GPT-4	N0	Unanimous	Majority of agents predicted N0 with high confidence, and clinical context supports low nodal burden.
2487	N0	GPT-4o	N0	Unanimous	Agent consensus strongly favors N0 with high confidence, and clinical context supports low nodal burden.
2487	N0	LLaMA-3			(not present in the LLaMA-3 Trust-OFF CSV)
2487	N0	Mistral-7B	N0	3–1 majority	3/4 agents (ResNet50, EffNetB3, ViT) agree on N0 with higher confidence; clinical summary consistent with low nodal burden.
3375	N0	GPT-4	N0	Unanimous	All agents unanimously predicted N0 with high confidence; the clinical context does not indicate a higher nodal burden.
3375	N0	GPT-4o	N0	Unanimous	Consensus supports N0 with high confidence, and clinical context does not suggest higher nodal burden.
3375	N0	LLaMA-3	N0	Unanimous	ResNet50 and VIT_B_16 agree on N0, with high confidence and no conflicting clinical context.
3375	N0	Mistral-7B			(not present in the Mistral-7B Trust-OFF CSV)
4199	N0	GPT-4	N+(1-2)	2–2/2–1–1 split	Majority favors N+(1–2) with higher confidence; clinical context aligns with labeled examples of N+(1–2).
4199	N0	GPT-4o	N+(1-2)	2–2/2–1–1 split	Agent agreement points to N+(1–2) with higher confidence; clinical context (low Ki67) supports lower nodal burden.
4199	N0	LLaMA-3	N+(1-2)	Unanimous	ResNet50 and EffNetB3 agree on N+(1–2) with high confidence.
4199	N0	Mistral-7B	N+(1-2)	2–2/2–1–1 split	Two agents (ResNet50, EffNetB3) support N+(1–2); other agents disagree; selected the 2-vote class.
5021	N0	GPT-4	N0	Unanimous	Majority of agents predicted N0 with high confidence, and clinical context supports low nodal burden.
5021	N0	GPT-4o	N0	Unanimous	Consensus favors N0 with high confidence, supported by clinical context and similar labeled examples.
5021	N0	LLaMA-3	N0	Unanimous	ResNet50 and VIT_B_16 agree on N0, with high confidence.
5021	N0	Mistral-7B	N0	3–1 majority	3/4 agents agree on N0 with higher confidence; clinical cues align with lower nodal burden.
6244	N+(1-2)	GPT-4	N+(>2)	Unanimous	Majority indicates N+(>2) with high confidence; clinical profile (ER+, PR+, HER2+) supports higher nodal burden.
6244	N+(1-2)	GPT-4o	N+(>2)	3–1 majority	Consensus leans to N+(>2) with high confidence; clinical context (HER2+, high Ki67) suggests higher nodal burden.
6244	N+(1-2)	LLaMA-3	N+(>2)	Unanimous	Strong agreement among agents, high confidence, and clinical context support N+(>2).
6244	N+(1-2)	Mistral-7B	N+(>2)	3–1 majority	3/4 agents support N+(>2) (ResNet50, EffNetB3, ViT); clinical summary supports higher nodal burden.

most structured reasonings by balancing consensus with trust scores. LLaMA-3 and GPT-4 offered reasonable, though shorter, rationales, while Mistral-7B often defaulted to the agent with marginally higher trust. Through these qualitative findings, we can deduce the fact that arbitration is not only numerically superior, but also explainable.

Although Swin-V2-S underperforms relative to the other experts, the arbitration framework is inherently robust to such disparities. The meta-learner downweights unreliable experts through low trust scores, ensuring that weaker models exert minimal influence on the final decision. Empirically, removing Swin-V2-S from the expert pool results in only marginal differences in arbitration accuracy, confirming that the LLM arbiter relies primarily on the more reliable experts while still benefiting from architectural diversity.

Although our framework supports multimodal prompting, the clinical metadata available in BCNB are low-dimensional and provide modest predictive value compared to image morphology. As a result, the clinical block contributed only small performance gains. This reflects dataset limitations rather than architectural constraints, and richer clinical or genomic features could yield larger multimodal benefits in future work. This distinction also reflects dataset availability: only BCNB provides clinical descriptors, whereas HMU contains imaging evidence alone, which naturally limits multimodal validation to the BCNB setting. While the performance differences between HMU and BCNB may appear to suggest a limited influence of clinical metadata, this reflects the differing nature of the tasks rather than the usefulness of multimodal prompting. HMU is an image-only, morphology-driven dataset, where improvements arise purely from arbitration and trust modeling. In BCNB, clinical descriptors provide complementary contextual signals that the LLM can leverage to resolve disagreements between experts or to stabilize decisions in borderline cases—an effect observed qualitatively in our case-level analyses. The multimodal block is therefore designed to be optional: it enhances reasoning when clinical cues are available, while the arbitration mechanism remains fully effective in image-only settings. As multimodal datasets evolve, the framework naturally supports inclusion of richer clinical context without modifying the arbitration process.

When we disable the meta learner's trust vector (Trust-OFF), the arbitrator no longer performs calibrated overrides in ambiguous cases, instead, it trends toward a consensus/confidence rule. This is reflected in the ablation studies over the BCNB dataset (see Table 6). For example, LLaMA-3 improves from 0.687→0.748 patch accuracy and from 0.519→0.747 patient level AUC with Trust-ON, while GPT-4o rises from 0.510→0.727 patch accuracy. These gains are consistent with an arbitration policy that can prioritize a reliable agent when the evidence and clinical cues justify it. The explanation content shifts accordingly (see Table 7) with Trust-OFF, "trust" drops to 0% of

reasons, while consensus/confidence language increases (e.g., GPT-4o consensus mentions approximately 52.4%) and models exit earlier at Unanimous/majority stages (e.g., LLaMA-3 Unanimous approximately 95.4%). Paired examples for the same BCNB patches (see Table 8) show the concrete cases that Trust-ON resolves via a clear trust margin (plus clinical alignment) flip to majority-driven decisions under Trust-OFF explaining the metric deltas without changing data or prompts. The proposed MELLMA framework addresses two critical barriers to AI adoption in pathology - robustness and trustworthiness. Traditional CNNs and Transformers often underperform in specific tissue subtypes or staining variations. By pooling multiple agents and arbitrating with an LLM, the system better handles outliers and ambiguous patches. This could potentially reduce false negatives, for example, in detecting nodal metastasis, a task with major implications for breast cancer treatment planning. Further, the ability of LLMs to produce transparent reasoning allows clinicians to inspect not only the outcome but also the rationale. This mirrors the human diagnostic workflow, where decisions are justified with references to morphology, biomarkers, or patient history. Such rationales are crucial for regulatory acceptance and clinical integration (Hager et al., 2024; Johri et al., 2025).

Although frontier LLMs such as GPT–4 and GPT–4o provide the most expressive clinical reasoning, we observed that lighter models (LLaMA–3–8B and Mistral–7B) occasionally achieved slightly higher accuracy on the BCNB dataset. BCNB prompts are highly structured and dominated by numerical signals (class, confidence, trust), with relatively limited free-text content. In such settings, performance is influenced more by strict rubric adherence and deterministic schema following than by broad reasoning capacity. Smaller models tended to follow the decision rubric and JSON output schema with lower variance, whereas frontier models sometimes produced more elaborate rationales or soft deviations from rubric wording. The resulting differences were small and within statistical variation, yet explain why lightweight models can surpass larger models in this specific, constrained prompting regime.

Although MELLMA introduces additional components beyond traditional ensembling, its computational cost is manageable and compatible with whole-slide workflows. Patch-level agent inference is parallelizable and efficiently batched on modern GPUs, and the trust-estimator MLP contributes negligible overhead. LLM arbitration represents the dominant per-patch cost, but the prompt size remains compact and the output schema is short, which keeps inference time stable even for lightweight 7–8B models. For whole-slide analysis, the framework does not require LLM arbitration on every patch. Standard tissue-detection or tumor-filtering modules can preselect a small subset of diagnostically relevant regions, reducing the number of LLM calls by one to two orders of magnitude in practice. Patient-level aggregation is deterministic and does not involve any LLM inference, making the final step computationally trivial. These design choices allow MELLMA to balance performance gains with computational feasibility. The arbitration provides measurable robustness to expert disagreement and produces structured, human-readable reasoning, which can justify the additional overhead in clinical decision-support contexts. On modern GPUs (e.g., NVIDIA V100/A100), forward passes for the individual expert models require only a few milliseconds per 224×224 patch (typically 3–7 ms depending on the architecture), and can be batched for high-throughput WSI processing. The meta-learner MLP adds negligible computational cost.

Although the LLM arbiter generally improves robustness to expert disagreement, there are several situations where arbitration may be unreliable. First, if all experts exhibit low confidence or inconsistent predictions, the aggregate trust vector becomes diffuse, and the LLM may receive insufficient signal to reach a reliable conclusion. Second, in highly ambiguous patches lacking diagnostic structures, the LLM may over-rely on subtle context or produce heuristic reasoning that is not clinically meaningful. Third, on borderline cases, the LLM may occasionally deviate from the decision rubric, particularly when expert outputs conflict strongly. To detect such failure cases, the framework provides several reliability cues: (i) the magnitude and entropy of the trust vector, (ii) disagreement patterns across experts, (iii) meta-learner confidence, and (iv) JSON-output validity checks to ensure rubric fidelity. These signals can be used to flag patches requiring manual review or alternative decision support, providing a practical layer of interpretability and error-awareness.

Despite promising results, our study has some limitations. First, the evaluation was conducted on two datasets, HMU-GC-HE-30K and BCNB, which, although complementary, do not capture the full diversity of cancer subtypes, histological variations, or staining protocols encountered in practice. Secondly, although arbitration improved quantitative performance, the choice of arbiter mattered. GPT-4o was the most consistent, but its proprietary nature and computational cost may limit deployment, while open-source models such as LLaMA-3, though competitive, still lag behind on some metrics. Thirdly, while LLMs provided interpretable rationales, these explanations were not always fully accurate or clinically grounded, underscoring the need for calibration mechanisms to verify the factuality of generated reasoning. Finally, running multiple CNN/Transformer agents plus an LLM arbitration layer introduces latency and computational cost, which could hinder real-time deployment. Future work can address these limitations in several ways. One promising direction is to extend arbitration from the patch level to slide level decision-making, enabling hierarchical reasoning across multiple scales of histological evidence. Additionally, domain specific fine-tuning of smaller, open source LLMs such as MedAlpaca (Han et al., 2023) or BioMistral (Labrak et al., 2024) may reproduce much of the interpretability of GPT-4o while reducing computational cost and improving accessibility (Huang et al., 2025a). Another way is to incorporate additional modalities beyond histology and basic clinical metadata, such as genomics or proteomics, which could allow arbitrators to synthesize richer evidence streams for improved decision making. However, future work should include additional agentic models (e.g., ConvNeXt (Woo et al., 2023) larger Swin Liu et al. (2021), and ViT (Dosovitskiy et al., 2020) variants) to further stress test arbitration, although early investigations suggest that the reported Trust-ON vs. Trust-OFF effects are stable across models. Finally, human-in-the-loop validation is critical. Future studies should test whether pathologists find LLM-generated rationales trustworthy and clinically useful, and whether arbitration reduces diagnostic errors in real practice settings. In addition, the arbitration framework is model-agnostic and can work with any visual model that provides task-specific predictions. Foundation models which are fine-tuned for specific diagnostic tasks and with support to provide such predictions, can be integrated into MELLMA without modifying the arbitration design.

6. Conclusion

We introduced a multi-expert LLM arbitration (MELLMA) framework in which a large language model (e.g., LLaMA-3, GPT-4o) consolidates diverse vision agents (e.g., ResNet-50, ViT-B/16) and clinical context through a structured prompt and a learned trust signal. The arbitrator not only reconciles conflicting predictions but also explains how a final decision is reached using consensus, trust margin, cluster proximity, or clinical tie-break, providing decisions that are auditable rather than opaque. By quantitatively calibrating agent confidence and injecting retrieval-based reasoning, the framework mitigates the known overconfidence and hallucination issues of standalone models in domain-specific tasks (Huang et al., 2025b; Tonmoy et al., 2024). Placed against the emerging literature on LLMs for medical imaging, our MELLMA framework addresses two recurring gaps, the need for controllable reasoning to mitigate ad hoc or hallucinated outputs, and the need for explanations aligned with clinical cues. The arbitration process provides insight into the expert who contributed to each decision and with what level of confidence. Clinicians can thus understand

why the system chose a particular output, improving transparency and potential user trust. This modular design means that newer expert models or data types can be integrated without fundamental changes, allowing the system to evolve with emerging diagnostic techniques.

CRediT authorship contribution statement

Shyam Sundar Debsarkar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **V.B. Surya Prasath:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Informed consent statement

There are no ethical implications regarding the public dataset.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets HMU-GC-HE-30K and BCNB are publically available at https://doi.org/10.6084/m9.figshare.25954813, https://bupt-ai-cz.github.io/BCNB/ respectively.

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